
LaBraM
VQ codebook
(2-stage)
original time domain
Mask time patches
raw signal, patches hidden
Transformer
encoder
Predict discrete code
codebook
predict: index #6
target: index in a learned codebook (trained stage 1)
Discrete token
VQ codebook
two-stage
Original time domain — a separately trained vector-quantizer maps each raw patch to a codebook index; the model predicts that index.
CBraMod
CSBrain
masked
reconstruction
original time domain
Mask time patches
raw signal, patches hidden
Transformer
encoder
Reconstruct original signal
target: original raw waveform
(includes task-irrelevant noise)
Raw waveform
reconstruct
signal
(noise included)
Original time domain — predict the raw microvolt waveform of masked patches; CSBrain uses the same target with cross-scale modelling.
Ours
codebook-free
spectral
frequency-band domain
Input EEG
raw waveform
learnable filterbank
5×Conv1d
init δθαβγ
δθαβγ + mask time
δ
θ
α
β
γ
band tokens; mask time patches
Transformer
encoder
Predict band power
δ
θ
α
β
γ
target: fixed-filterbank log band-power (no codebook)
Band power
spectral MSE vs.
fixed target
+ 0.5·recon +
SIGReg
Frequency-band domain — a learnable filterbank (5 depthwise Conv1d per channel, initialised at δθαβγ centres) maps each patch into bands; we then predict the masked patches' log band-power against a fixed filterbank target. The domain (bands, not raw) and the target (power, not waveform) are what differ; the mask axis (time) is shared.
δ
θ
α
β
γ
masked patch
prediction
target